In [1]:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression, LogisticRe
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, mean_absolute

# Load data
df = pd.read_csv("temperatures.csv")

print(df.head())
print("\nShape:", df.shape)
print("\nColumns:", df.columns)

# Select month and temperature
X = df[["YEAR"]]
y = df["ANNUAL"]

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

# Linear Regression
linear_model = LinearRegression()
linear_model.fit(X_train, y_train)

y_pred = linear_model.predict(X_test)

print("\nLinear Regression")
print("MSE:", mean_squared_error(y_test, y_pred))
print("MAE:", mean_absolute_error(y_test, y_pred))
print("R-Square:", r2_score(y_test, y_pred))

# Predict temperature
print("\nPredicted Temperatures:")
print(y_pred)

# Logistic Regression
# Create temperature categories
df["Temperature_Category"] = pd.cut(
    df["ANNUAL"],
    bins=[-float("inf"), 25, 30, float("inf")],
    labels=[0, 1, 2]
)

X = df[["YEAR"]]
y = df["Temperature_Category"].astype(int)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

logistic_model = LogisticRegression(max_iter=1000)
logistic_model.fit(X_train, y_train)
```

```
logistic_pred = logistic_model.predict(X_test)

print("\nLogistic Regression")
print("Predicted Categories:")
print(logistic_pred)

# Visualization
plt.scatter(df["YEAR"], df["ANNUAL"])
plt.plot(
    X_test["YEAR"],
    y_pred,
    linewidth=2
)
plt.xlabel("Year")
plt.ylabel("Temperature (°C)")
plt.title("Linear Regression - Temperature")
plt.show()
```

```
   YEAR    JAN    FEB    MAR    APR    MAY    JUN    JUL    A
UG    SEP    OCT  \
0  1901  22.40  24.14  29.07  31.91  33.41  33.18  31.21  30.
39  30.47  29.97
1  1902  24.93  26.58  29.77  31.78  33.73  32.91  30.92  30.
73  29.80  29.12
2  1903  23.44  25.03  27.83  31.39  32.91  33.00  31.34  29.
98  29.85  29.04
3  1904  22.50  24.73  28.21  32.02  32.64  32.07  30.36  30.
09  30.04  29.20
4  1905  22.00  22.83  26.68  30.01  33.32  33.25  31.44  30.
68  30.12  30.67

     NOV    DEC  ANNUAL  JAN-FEB  MAR-MAY  JUN-SEP  OCT-DEC
0  27.31  24.49   28.96    23.27    31.46    31.27    27.25
1  26.31  24.04   29.22    25.75    31.76    31.09    26.49
2  26.08  23.65   28.47    24.24    30.71    30.92    26.26
3  26.36  23.63   28.49    23.62    30.95    30.66    26.40
4  27.52  23.82   28.30    22.25    30.00    31.33    26.57

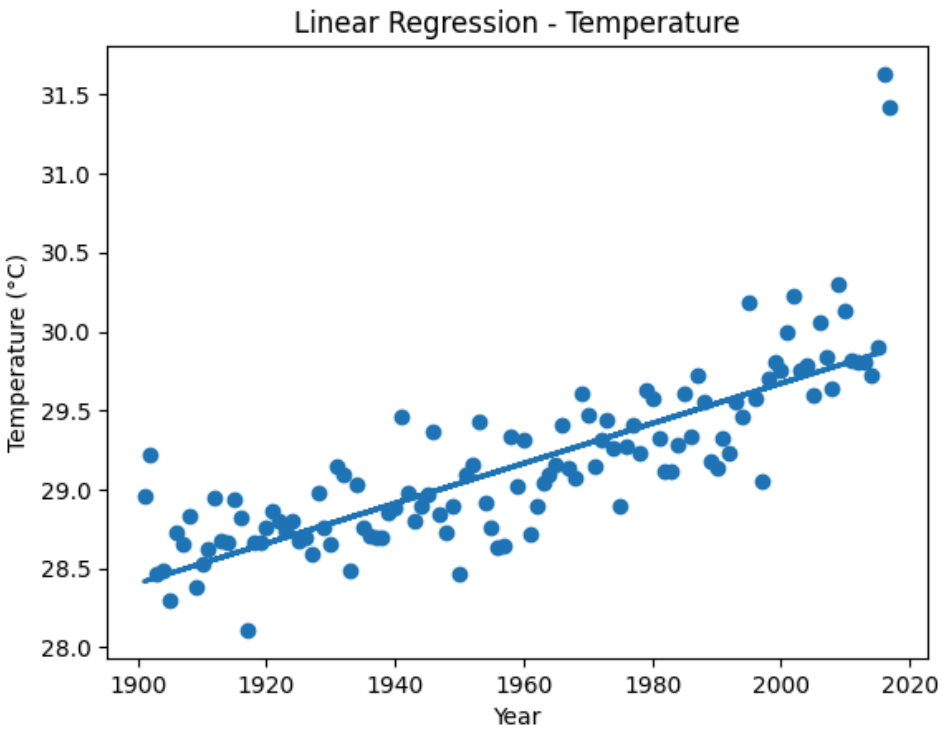
Shape: (117, 18)

Columns: Index(['YEAR', 'JAN', 'FEB', 'MAR', 'APR', 'MAY', 'J
UN', 'JUL', 'AUG', 'SEP',
       'OCT', 'NOV', 'DEC', 'ANNUAL', 'JAN-FEB', 'MAR-MAY',
'JUN-SEP',
       'OCT-DEC'],
      dtype='object')

Linear Regression
MSE: 0.17794903368181422
MAE: 0.2667162919914695
R-Square: 0.5820853486434291

Predicted Temperatures:
[28.97496203 28.46859749 29.08889405 28.9496438  28.54455217
29.49398568
 29.3294172  29.6079177  28.87368912 28.55721129 28.92432558
28.74709799
 29.62057681 29.73450883 28.64582508 28.41796104 29.20282607
29.83578174
 29.38005366 29.69653149 29.27878075 29.54462213 29.87375908
28.5698704
 28.60784774 29.29143986 28.81039355 28.72177976 29.12687139
28.69646153]

Logistic Regression
Predicted Categories:
[1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1]
```


Linear Regression - Temperature
Temperature (°C)
31.5
31.0
30.5
30.0
29.5
29.0
28.5
28.0
1900
1920
1940
1960
1980
2000
2020
Year